Ordinary Differential Equations

Lecture Notes

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1 Introduction to Modern Developments

The field of differential equations is evolving quickly, but the language of this lecture remains ordinary differential equations (ODEs). We keep three strands together: (i) deep models in the continuous–depth limit (Neural ODEs) and their training; (ii) data–driven discovery of right–hand sides with explicit control of complexity and noise; and (iii) networked and multiscale systems formulated directly as ODEs. The emphasis is on precise formulations, assumptions, and links back to existence–uniqueness, stability, and numerical accuracy from earlier lectures.

Large, irregular datasets can now be fit with ODE models by differentiating through solvers; high–dimensional coupled systems on graphs are analyzed spectrally without leaving the ODE setting; and singular perturbations reduce fast–slow models to lower–dimensional ODEs that retain essential behavior.

1.1 Historical Context and Motivation

Classical analysis supplied closed forms in special families and qualitative structure; the twentieth century added robust numerics. Two shifts now reshape practice. First, automatic differentiation exposes gradients through ODE solvers, enabling end–to–end training of models whose evolution is governed by ODEs. Second, data–rich settings motivate identifying right–hand sides from measurements with parsimony. These do not replace the analytic core: Lipschitz conditions still guarantee well–posedness; linearization and Lyapunov arguments still decide stability; stiffness still dictates solver choice. The contemporary loop is: propose an ODE, learn unknown parts if needed, validate on withheld trajectories, and iterate.

2 Neural Ordinary Differential Equations

2.1 Conceptual foundation

A residual network with step size Δ updates

$$h_{k+1} = h_k + \Delta\, f(h_k, t_k, \theta).$$

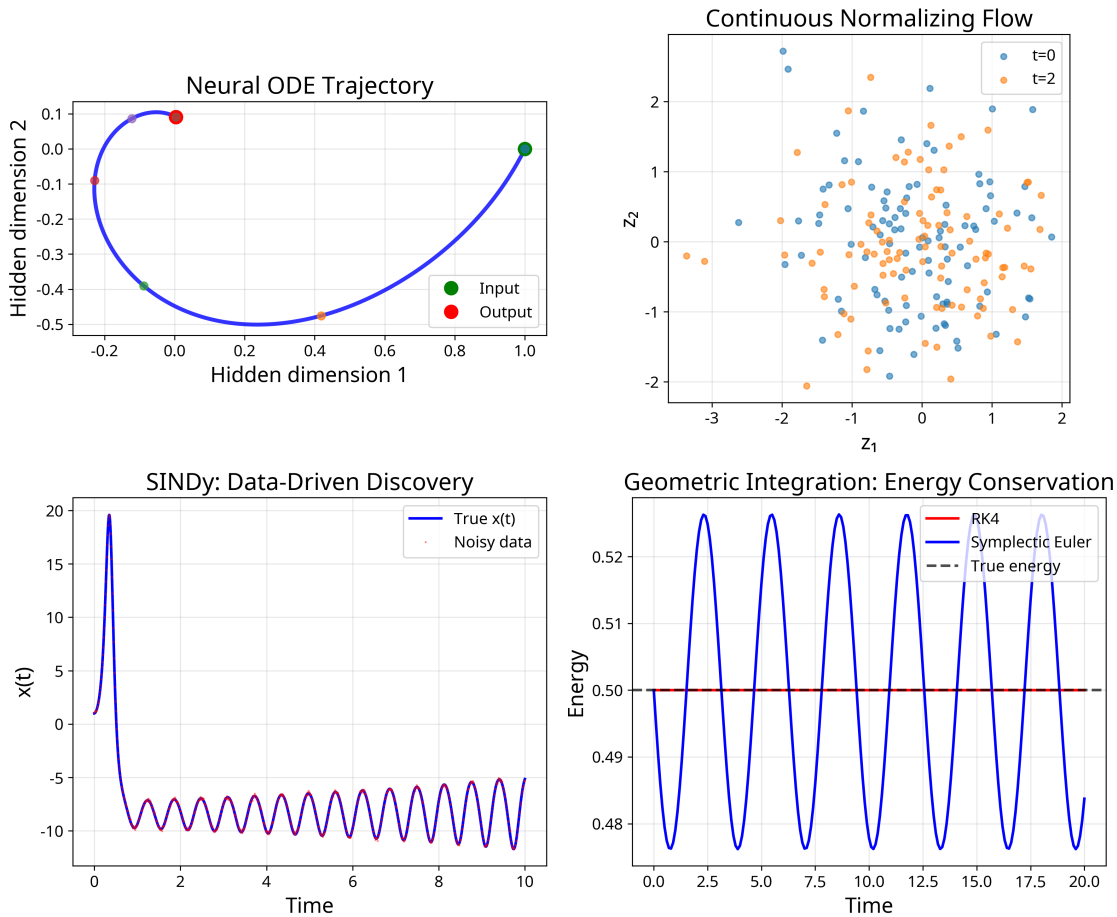

Figure 1: Within the ODE framework: residual networks in the continuous–depth limit and adjoint training; sparse, noise–robust discovery of right–hand sides; dynamics on graphs via Laplacians and phase coupling; and fast–slow reduction for multiscale ODEs.

Letting $\Delta \to 0$ on a fixed horizon gives the initial–value problem

$$\frac{dh}{dt} = f(h(t), t, \theta), \qquad h(0) = x, \qquad \hat{y} = g\big(h(T)\big). \tag{1}$$

If f is locally Lipschitz in h (uniformly in t), (1) is well posed and depends continuously on (x, θ). The learned flow $\Phi_t^\theta : x \mapsto h(t)$ is a diffeomorphism when f is C^1 and trajectories avoid blow–up. Volume change is controlled by the divergence:

$$\frac{d}{dt} \log \det D\Phi_t^\theta(x) = \operatorname{tr}\big(\partial_h f(h(t), t, \theta)\big).$$

Divergence–free f produce volume–preserving flows.

Example (linear Neural ODE). Let $f(h, t, \theta) = Wh$ with parameter $W \in \mathbb{R}^{d\times d}$. Then $h(t) = e^{tW}x$ and $\hat{y} = g(e^{TW}x)$. A residual net with small step Δ and block $h \mapsto h + \Delta Wh$ approximates e^{TW} by $(I + \Delta W)^{T/\Delta} \to e^{TW}$ as $\Delta \to 0$. This makes the continuous–depth view explicit in a case with closed form.

2.2 Training by adjoint sensitivity

For a terminal loss $L = L(h(T))$, define $a(t) = \partial L/\partial h(t)$. A standard variational argument yields

$$\frac{da}{dt} = -\big(\partial_h f(h(t), t, \theta)\big)^\top a(t), \qquad a(T) = \partial_h L\big(h(T)\big), \tag{2}$$

$$\frac{\partial L}{\partial \theta} = \int_0^T a(t)^\top \, \partial_\theta f\big(h(t), t, \theta\big)\, dt, \qquad \frac{\partial L}{\partial x} = a(0). \tag{3}$$

The adjoint equation (2) describes the reverse-time sensitivity of the loss with respect to the state $h(t)$, while (3) gives the gradient of the loss with respect to the parameters θ. In practice the gradient depends on discretization. With a one–step Runge–Kutta method

$$h_{n+1} = h_n + h \sum_{i=1}^{s} b_i \, k_i, \qquad k_i = f\big(h_n + h \sum_{j=1}^{s} a_{ij} k_j,\ t_n + c_i h,\ \theta\big),$$

the discrete adjoint (reverse-time sensitivity) λ_n at step n satisfies

$$\lambda_n = \Big(I + h \sum_{i=1}^{s} b_i \, \partial_h f_i \, \big(I + h \sum_j a_{ij} \partial_h f_j\big)^{-1}\Big)^\top \lambda_{n+1}$$

(implicit in general) aligns the trained objective with the computed trajectory ("discretize-then-optimize"). For stiff f, this route is typically more stable than discretizing (2) after the fact.

2.3 Structured and augmented models

Expressivity can be increased by augmenting the state:

$$\frac{d}{dt}\begin{pmatrix} h \\ a \end{pmatrix} = F_\theta\left(\begin{pmatrix} h \\ a \end{pmatrix}, t\right).$$

When invariants are known, restrict f to a structured class. Example: energy–conserving forms $f(h,t) = J\nabla H(h,t)$ with $J^\top = -J$ preserve H exactly in the continuous limit; parameterizing H rather than f respects this structure during training.

2.4 Continuous normalizing flows

An invertible flow $\dot{h} = f_\theta(h,t)$ transforms densities by Liouville's formula. Intuitively, $\mathrm{tr}(\partial_h f_\theta)$ is the instantaneous volume change (divergence) induced by the vector field, so accumulating it along the path adjusts log-density:

$$\frac{d}{dt} \log p(h(t)) = -\mathrm{tr}\,\big(\partial_h f_\theta(h(t), t)\big),$$

so

$$\log p\big(h(T)\big) = \log p\big(h(0)\big) - \int_0^T \mathrm{tr}\,(\partial_h f_\theta)\, dt.$$

Training uses an augmented ODE for $(h,\ s)$ with $s' = -\mathrm{tr}(\partial_h f_\theta)$, keeping everything in the ODE framework.

Example (1D CNF). If $h' = \alpha(t)\,h$ on $[0,T]$ with known α, then $h(T) = h(0)\,\exp(\int_0^T \alpha)$ and the log-density transforms as

$$\log p\big(h(T)\big) = \log p\big(h(0)\big) - \int_0^T \alpha(t)\,dt,$$

since $\mathrm{tr}(\partial_h f) = \partial_h(\alpha h) = \alpha$. Thus positive (negative) average α expands (contracts) mass.

3 Data–Driven Discovery of Differential Equations

3.1 Sparse identification (SINDy)

Assume $\dot{x} = f(x)$ and that f is sparse in a library $\Theta(x) = [\phi_1(x), \ldots, \phi_m(x)]$. With samples $X = [x(t_1), \ldots, x(t_N)]^\top$ and $\dot{X} = [\dot{x}(t_1), \ldots, \dot{x}(t_N)]^\top$, pose

$$\dot{X} = \Theta(X)\,\Xi + E, \tag{4}$$

and promote sparsity in Ξ either by a convex surrogate

$$\min_{\Xi}\ \|\Theta(X)\Xi - \dot{X}\|_F^2 + \lambda\|\Xi\|_{1,1},$$

or by *sequential thresholded least squares* (STLS):

$$\Xi^{(k+1)} = \mathcal{T}_\tau\Big(\big(\Theta^\top\Theta\big)^{-1}\Theta^\top\dot{X}\Big), \quad \mathcal{T}_\tau(\beta_j) = \beta_j\mathbf{1}\{|\beta_j| > \tau\},$$

iterated until the active set stabilizes. Validation is done on withheld *trajectories*, not interleaved points, to test predictive power. Persistently exciting inputs and multiple initial conditions improve identifiability.

Example (recover a cubic from data). Generate samples from $x' = 2x - x^3$ with several initial conditions and small additive noise. Build a library $\Theta = [1, x, x^2, x^3]$ and estimate $\dot{x}$ either by smoothing then differentiating or by the weak form. Solving (4) with sparsity (e.g., STLS) should select the columns for x and x^3 with coefficients close to $(2, -1)$, while zeroing the others. Validate by simulating the identified ODE on unseen initial conditions.

3.2 Noise–robust weak formulation

Finite differences amplify noise. A weak form avoids pointwise differentiation: multiply by smooth test functions ψ_k and integrate by parts,

$$\int_0^T \psi_k\,\dot{x}\,dt = -\int_0^T \psi_k'\,x\,dt, \qquad \int_0^T \psi_k\,f(x(t))\,dt \approx \sum_\ell w_{k\ell}\,\Theta(x(t_\ell)),$$

yielding a linear system in the coefficients of Θ with far better conditioning. Bootstrap resampling of trajectories produces an ensemble of fits; terms that persist across the ensemble are retained, providing uncertainty on structure.

3.3 Residual–penalty fitting for partially known models

When f is known up to parameters θ or a small correction, fit a smooth surrogate u by balancing data and ODE residual at collocation nodes $\{\tau_j\}$,

$$\mathcal{J}(u,\theta) = \sum_i \|u(t_i) - y_i\|^2 + \lambda \sum_j \left\| \tfrac{d}{dt} u(\tau_j) - f\big(u(\tau_j), \tau_j, \theta\big) \right\|^2,$$

with derivatives obtained by automatic differentiation through the interpolant (e.g., cubic splines). This keeps mechanistic structure while letting data tune unknowns.

4 Complex Networks and Graph Dynamics

4.1 Consensus via Laplacian flows

Consensus means a group of agents with scalar states $x_i(t)$ agreeing on a common value through local interactions. For an undirected graph with adjacency matrix A and degrees $d_i = \sum_j A_{ij}$, the graph Laplacian is $L = D - A$ with $D = \mathrm{diag}(d_i)$. The canonical continuous-time consensus dynamics nudges each agent toward the average of its neighbors; with coupling $\sigma > 0$ this reads

$$\dot{x} = -\sigma L x, \qquad x \in \mathbb{R}^n.$$

This flow preserves the average $\bar{x} = \frac{1}{n}\mathbf{1}^\top x$ because $L\mathbf{1} = 0$; hence the only possible agreement point is $\bar{x}\,\mathbf{1}$. A standard Lyapunov function $V(x) = \frac{1}{2}x^\top L x$ decreases along trajectories:

$$\dot{V} = x^\top L \dot{x} = -\sigma x^\top L^2 x \le -2\sigma\lambda_2(L)\, V,$$

so

$$V(t) \le V(0)\, e^{-2\sigma\lambda_2(L)t}, \qquad \|x(t) - \bar{x}\,\mathbf{1}\| \le C\, e^{-\sigma\lambda_2(L)t} \|x(0) - \bar{x}\,\mathbf{1}\|.$$

Rates are governed spectrally by the algebraic connectivity $\lambda_2(L)$.

Example (three agents on a path). For nodes 1–2–3 with unit edges and $\sigma = 1$, the Laplacian is $\left(\begin{smallmatrix} 1 & -1 & 0 \\ -1 & 2 & -1 \\ 0 & -1 & 1 \end{smallmatrix}\right)$. With $x(0) = (1,3,5)^\top$, the invariant average is $\bar{x} = 3$. Numerical integration shows $x(t) \to (3,3,3)^\top$ and the disagreement $\|x(t) - \bar{x}\,\mathbf{1}\|$ decays approximately like $e^{-\lambda_2 t}$, where $\lambda_2 > 0$ is the second-smallest eigenvalue of L.

Two-agent example. For two nodes coupled by one edge, $L = \left(\begin{smallmatrix} 1 & -1 \\ -1 & 1 \end{smallmatrix}\right)$ and

$$\dot{x}_1 = -\sigma(x_1 - x_2), \qquad \dot{x}_2 = -\sigma(x_2 - x_1).$$

Define the disagreement $e = x_1 - x_2$. Then $\dot{e} = -2\sigma e$ so $e(t) = e(0)e^{-2\sigma t}$. Meanwhile the average $\bar{x} = (x_1 + x_2)/2$ is constant. Thus both agents converge exponentially to the common value $\bar{x}$.
Interpretation and exercise. The Laplacian flow $\dot{x} = -\sigma L x$ preserves the average and contracts the disagreement $x - \bar{x}\mathbf{1}$. On connected graphs, the slowest mode decays at rate $\sigma\lambda_2(L)$. Using Fig. 2, (i) verify numerically that $\log\|x - \bar{x}\mathbf{1}\|$ is approximately linear and estimate its slope; (ii) change the topology (path $\to$ ring $\to$ complete) and compare the estimated slope with $\lambda_2(L)$; (iii) explore the effect of rescaling σ.

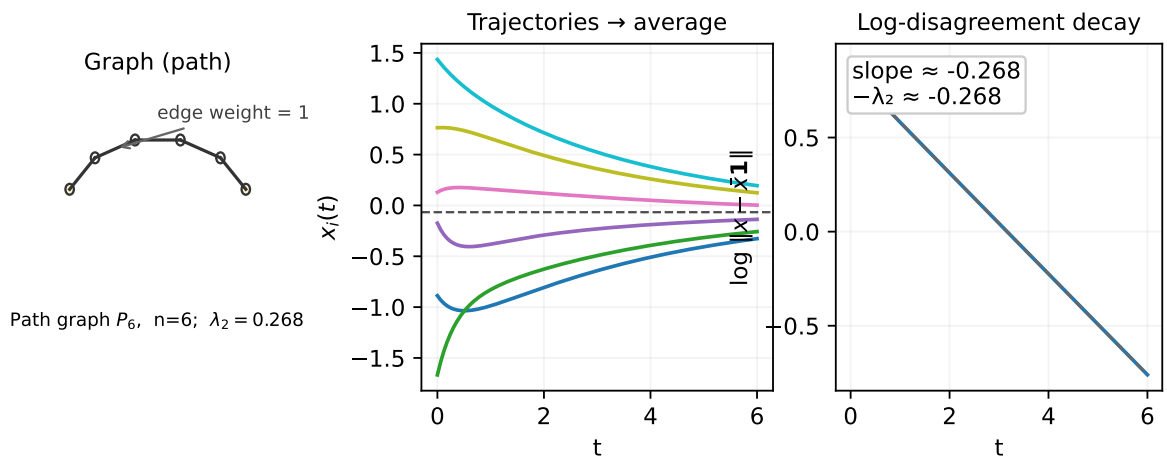

Figure 2: Exercise figure for consensus on a path graph ($n = 6$). Left: the graph. Middle: state trajectories $x_i(t)$ converging to the invariant average $\bar{x}$. Right: the log-disagreement $\log \|x - \bar{x}\mathbf{1}\|$ versus time, which decays approximately linearly with slope $\approx -\lambda_2(L)$, illustrating the spectral rate.

4.2 Phase synchronization in the simplest case

For two oscillators, let $\Delta = \omega_1 - \omega_2$ be the natural frequency mismatch and $K > 0$ the coupling strength; the phase difference ϕ then obeys the scalar ODE (a reduced consensus of phases)

$$\dot{\phi} = \Delta - K \sin \phi, \qquad \phi = \theta_1 - \theta_2.$$

Equilibria satisfy $\sin \phi^* = \Delta/K$, which exist iff $|\Delta| \leq K$. Linearizing gives $\dot{\eta} = -(K \cos \phi^*)\eta$, so the locked state with $\cos \phi^* > 0$ is asymptotically stable. Larger networks reduce along Laplacian eigenmodes; thresholds and rates are again spectral.

5 Multiscale Methods

5.1 Tikhonov–Fenichel reduction

We partition variables into slow x and fast y with a small parameter $\varepsilon > 0$ that sets the time-scale separation:

$$x' = f(x, y), \qquad \varepsilon y' = g(x, y), \qquad 0 < \varepsilon \ll 1.$$

The *critical manifold* $M = \{(x, y) : g(x, y) = 0\}$ is locally the graph $y = h_0(x)$ if $\partial_y g(x, h_0(x))$ is invertible. If the fast subsystem $y' = \frac{1}{\varepsilon} g(x, y)$ has M normally hyperbolic and attracting, then solutions starting near M satisfy

$$\|(x(t), y(t)) - (\bar{x}(t), h_0(\bar{x}(t)))\| \leq C\varepsilon$$

for $t = O(1)$ after an initial layer of length $O(\varepsilon \log \varepsilon^{-1})$, where $\bar{x}' = f(\bar{x}, h_0(\bar{x}))$. A first–order correction $h_1(x)$ solves

$$\partial_y g(x, h_0(x))\, h_1(x) = -\, \partial_x g(x, h_0(x))\, f(x, h_0(x)),$$

giving $y = h_0(x) + \varepsilon h_1(x) + O(\varepsilon^2)$ and an $O(\varepsilon)$–accurate reduced model.

Example (Michaelis–Menten reduction). Let $x = S$ (substrate) be slow and $y = C$ (complex) be fast. With enzyme total E_0 and rates k_1, k_{-1}, k_2, write

$$S' = -k_1 S(E_0 - C) + k_{-1}C, \qquad \varepsilon C' = k_1 S(E_0 - C) - (k_{-1} + k_2)C,$$

with small $\varepsilon > 0$ marking fast complex dynamics. The critical manifold solves $0 = k_1 S(E_0 - C) - (k_{-1} + k_2)C$, so $C = \dfrac{E_0 S}{K_M + S}$ with $K_M = (k_{-1} + k_2)/k_1$. Substituting gives the reduced scalar ODE

$$S' = -\frac{k_2 E_0 S}{K_M + S},$$

accurate up to $O(\varepsilon)$ after a short initial layer.

5.2 Averaging for weakly perturbed oscillators

For $z' = Az + \varepsilon F(z, t)$ with A skew–symmetric (purely imaginary spectrum), write $z = re^{i\theta}$ and average over the fast phase:

$$\bar{r}' = \varepsilon \frac{1}{2\pi} \int_0^{2\pi} \Re\big(e^{-i\theta} F(re^{i\theta}, t)\big)\, d\theta, \qquad \bar{\theta}' = \omega + O(\varepsilon).$$

Then $\|z(t) - \bar{z}(t)\| = O(\varepsilon)$ for $t = O(\varepsilon^{-1})$. Averaging exposes slow envelopes and bifurcations in a reduced ODE.

Example (near-resonant phase averaging). Consider phases with weak coupling to a periodic drive: $\theta' = \omega + \varepsilon A \sin(\Omega t - \theta)$. Let $\psi = \theta - \Omega t$. Then $\psi' = \Delta - \varepsilon A \sin\psi$ with $\Delta = \omega - \Omega$. The averaged dynamics is already autonomous: for $|\Delta| < \varepsilon A$ there is a stable locked state solving $\sin\psi^* = \Delta/(\varepsilon A)$; otherwise the phase drifts with mean rate $\approx \operatorname{sign}(\Delta)\sqrt{\Delta^2 - (\varepsilon A)^2}$.

6 Learning, Estimation, and Control with ODEs

6.1 Differentiable programming and parameter estimation

Automatic differentiation through ODE solvers enables maximum–likelihood or least–squares estimation for mechanistic ODEs. Alignment of training and evaluation requires consistent interpolation, event handling, and tolerances. Regularizing $\|\partial_h f\|$ or constraining Lipschitz constants curbs exploding sensitivities and implicit stiffness.

6.2 Optimal control viewpoint

With dynamics $\dot{x} = f(x, u, \theta)$ and cost

$$J = \Phi(x(T)) + \int_0^T \ell(x, u)\, dt,$$

Pontryagin's principle introduces the Hamiltonian $H(x, p, u) = \ell(x, u) + p^\top f(x, u)$ and yields

$$\dot{x} = \partial_p H, \qquad \dot{p} = -\partial_x H, \qquad \partial_u H = 0,$$

a two–point boundary–value ODE system with $x(0) = x_0$, $p(T) = \partial_x \Phi(x(T))$. Shooting and adjoint gradients put this on the same computational footing as Neural ODE training.

7 Computational Considerations

Stability and stiffness. For a one–step method applied to $y' = \lambda y$, the amplification factor is $R(z)$ with $z = \lambda h$. Absolute stability requires $|R(z)| < 1$ for relevant z. Explicit Euler has $R(z) = 1 + z$ (stable if $z \in (-2, 0)$); Backward Euler has $R(z) = 1/(1 - z)$ (A– and L–stable). Stiffness $\big(\max|\Re\lambda| \gg \min|\Re\lambda|\big)$ forces explicit steps $h \lesssim 2/|\lambda_{\text{stiff}}|$, motivating implicit or stabilized schemes.

Accuracy and adaptivity. For an order–p method, local error is $O(h^{p+1})$. Embedded pairs estimate error e and update h by

$$h_{\text{new}} = h\left(\frac{\text{tol}}{|e|}\right)^{1/(p+1)} \rho, \qquad \rho \in (0, 1).$$

Reproducibility. Report solver, tolerances, and random seeds; checkpoint states. When training Neural ODEs, differentiating the *discrete* scheme aligns optimization with computation, especially under adaptive stepping.

Exercises

Exercise 1

Let $h'(t) = \theta\, h(t)^3$, $h(0) = h_0 \neq 0$, and $L = \frac{1}{2}\big(h(T) - y\big)^2$.

1. Solve the IVP and write $h(T)$ in terms of (h_0, θ, T).
2. Derive the adjoint $a'(t) = -(\partial_h f)\, a$, $a(T) = h(T) - y$, and show
$$\frac{\partial L}{\partial \theta} = \int_0^T a(t)\, h(t)^3\, dt, \qquad \frac{\partial L}{\partial h_0} = a(0).$$
3. Explain how bounding $\|\partial_h f\| = |3\theta h(t)^2|$ controls the growth of $a(t)$ and stabilizes training.

Exercise 2

Consider $h'(t) = \theta h(t)$, $h(0) = h_0$, $L = \frac{1}{2}(h(T) - y)^2$.

1. Compute $\partial L/\partial\theta$ using the continuous adjoint.
2. With Forward Euler of step Δ ($N = T/\Delta$ steps), $h_{n+1} = (1 + \Delta\theta)h_n$. Backpropagate through the recurrence to obtain $dL_N/d\theta$ and show $dL_N/d\theta = \partial L/\partial\theta + O(\Delta)$; identify the leading error term.

Exercise 3

Let $h'(t) = f_\theta(h, t)$ and define $s'(t) = -\operatorname{tr} \partial_h f_\theta(h(t), t)$, $s(0) = 0$.

1. Prove $\log p(h(T)) = \log p(h(0)) - s(T)$.
2. For $f_\theta(h, t) = Ah$ with constant A, compute $s(T)$ and verify $\det D\Phi_T = e^{T \operatorname{tr}(A)}$.
3. Take $A = \left(\begin{smallmatrix} 0 & -1 \\ 1 & 0 \end{smallmatrix}\right)$. Determine $s(T)$ and discuss the effect on the transported density.

Exercise 4

Assume $x'(t) = ax + bx^3$ and (clean) samples $(x_i, \dot{x}_i) = (1, 1.4), (2, 6.5), (-1, -1.2), (0.5, 0.4)$.

1. Build $\Theta(x) = [\,x \;\; x^3\,]$ and solve the least squares normal equations for $(\hat{a}, \hat{b})$. If $|\hat{b}| < 0.1$, set it to 0 and report the model.

2. Suppose $\dot{x}$ is noisy and direct differencing fails. With $\psi_1(t) = 1$, $\psi_2(t) = 1 - 2t/T$, derive $\int_0^T \psi_k x'\,dt = -\int_0^T \psi_k' x\,dt + [\psi_k x]_0^T$ and write the linear system that links $\big(\int \psi_k x\,dt,\ \int \psi_k x^3\,dt\big)$ to (a, b).

Exercise 5

For $\dot{x} = -\sigma L x$ on a connected undirected graph,

1. Show the average is invariant and that $V(x) = \frac{1}{2} x^\top L x$ satisfies $\dot{V} \le -2\sigma\lambda_2(L)\,V$; deduce a bound on $\|x(t) - \bar{x}\,\mathbf{1}\|$.

2. Compare K_n (complete) vs. P_n (path): use $\lambda_2(K_n) = n$ and $\lambda_2(P_n) = 2(1 - \cos(\pi/n))$ to discuss scaling with n and implications for fast consensus.

Exercise 6

$$x' = -x + y, \qquad \varepsilon\, y' = -y + x^2, \qquad 0 < \varepsilon \ll 1.$$

1. Find the critical manifold $M:\ g(x, y) = 0$ and the reduced slow flow $x' = f\big(x, h_0(x)\big)$.

2. Compute $h_1(x)$ from $\partial_y g(x, h_0)\, h_1(x) = -\partial_x g(x, h_0)\, f\big(x, h_0(x)\big)$ and form $y \approx h_0(x) + \varepsilon h_1(x)$.

3. Locate equilibria of reduced and full systems and explain why stability agrees for small ε; estimate the $O(\varepsilon \log \varepsilon^{-1})$ initial layer.